MEMBRANES

Module map

- Module 05: Membranes
- Connected lab: lab-05_viewing-life.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 05

Membranes

1. Phospholipid bilayer structure
2. Selective permeability
3. Passive transport
4. Active transport
5. Osmosis and cell environments

Lab evidence

lab-05_viewing-life.md

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Learning objectives

- Describe how phospholipid structure creates a bilayer.
- Predict which molecules cross membranes easily.
- Compare passive and active transport.
- Use concentration gradients to explain movement across membranes.
- Predict osmosis outcomes in hypotonic, isotonic, and hypertonic conditions.

OBJECTIVE 1
Describe how phospholipid structure creates a bilayer.

OBJECTIVE 2
Predict which molecules cross membranes easily.

OBJECTIVE 3
Compare passive and active transport.

OBJECTIVE 4
Use concentration gradients to explain movement across membranes.

OBJECTIVE 5
Predict osmosis outcomes in hypotonic, isotonic, and hypertonic conditions.

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Topic sequence

- Phospholipid bilayer structure: The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.
- Selective permeability: Selective permeability controls which materials cross easily and which require help.
- Passive transport: Diffusion, osmosis, and facilitated diffusion move substances down gradients.
- Active transport: Active transport uses energy to move substances against gradients.
- Osmosis and cell environments: Tonicity predicts water movement and cell volume changes.

01

Phospholipid bilayer structure

The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.

02

Selective permeability

Selective permeability controls which materials cross easily and which require help.

03

Passive transport

Diffusion, osmosis, and facilitated diffusion move substances down gradients.

04

Active transport

Active transport uses energy to move substances against gradients.

05

Osmosis and cell environments

Tonicity predicts water movement and cell volume changes.

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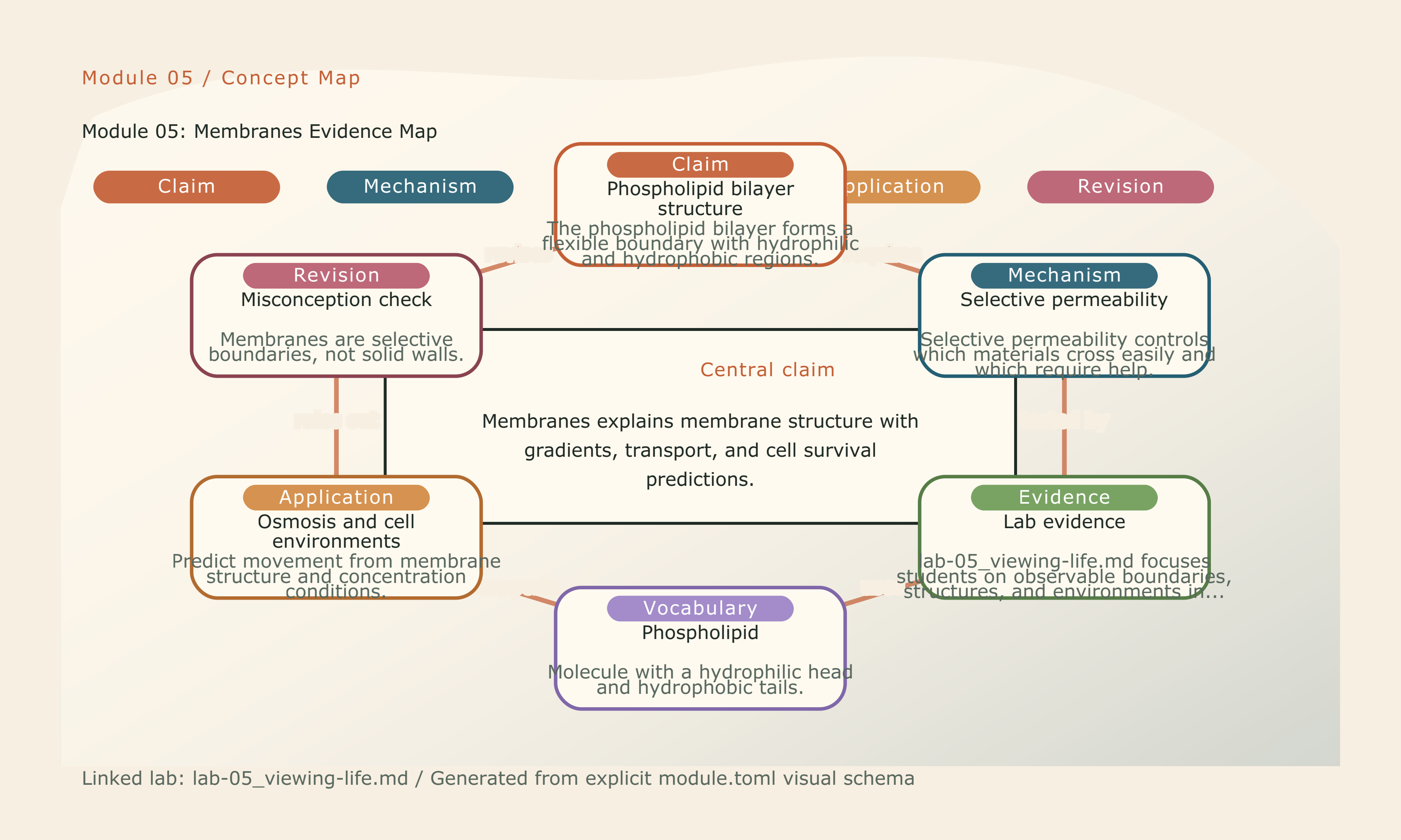
Module 05: Membranes Evidence Map

- Module focus: Membranes
- Central claim: Membranes explains membrane structure with gradients, transport, and cell survival predictions.
- Key nodes to connect: Phospholipid bilayer structure, Selective permeability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-05_viewing-life.md supplies an evidence surface for the map.



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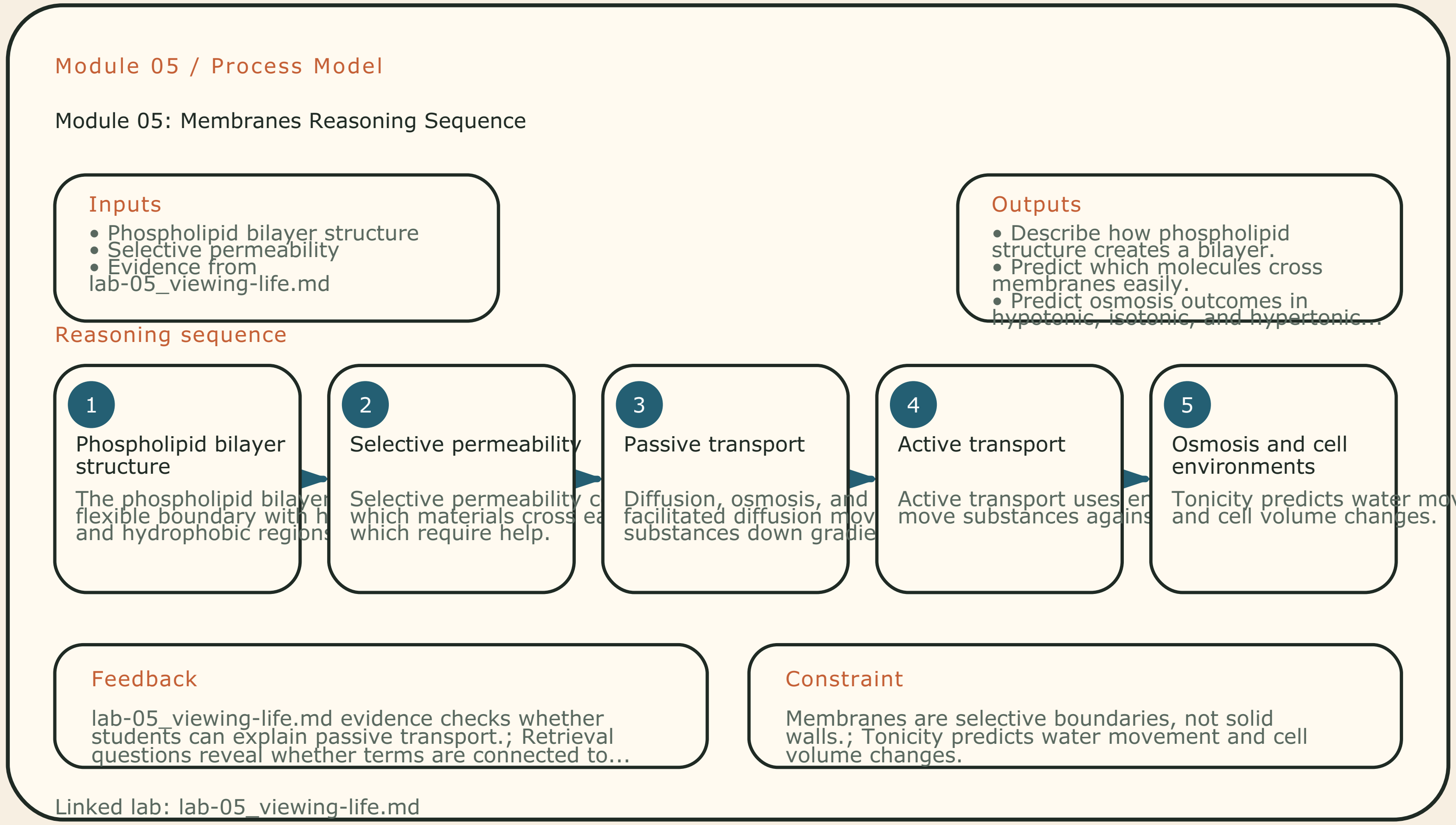
Module 05: Membranes Reasoning Sequence

- Module focus: Membranes
- Trace the model: Phospholipid bilayer structure -> Selective permeability -> Passive transport -> Active transport
- Inputs become outputs: Phospholipid bilayer structure, Selective permeability -> Describe how phospholipid structure creates a bilayer., Predict which molecules cross membranes easily.
- Feedback check: lab-05_viewing-life.md evidence checks whether students can explain passive transport.
- Lab connection: lab-05_viewing-life.md gives students a process to observe or test.



MEMBRANES

Terms as evidence handles

- Phospholipid: Molecule with a hydrophilic head and hydrophobic tails.
- Bilayer: Double layer of phospholipids forming a membrane.
- Selective permeability: Property of allowing some substances through more easily than others.
- Diffusion: Net movement from high to low concentration.
- Osmosis: Diffusion of water across a selectively permeable membrane.
- Facilitated diffusion: Passive transport through membrane proteins.

PHOSPHOLIPID
Molecule with a hydrophilic head and hydrophobic tails.

BILAYER
Double layer of phospholipids forming a membrane.

SELECTIVE PERMEABILITY
Property of allowing some substances through more easily than others.

DIFFUSION
Net movement from high to low concentration.

OSMOSIS
Diffusion of water across a selectively permeable membrane.

FACILITATED DIFFUSION
Passive transport through membrane proteins.

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Lab connection

- Lab file: lab-05_viewing-life.md
- Lab evidence should test or illustrate: Passive transport
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Phospholipid bilayer structure

EVIDENCE
lab-05_viewing-life.md

CLAIM
Describe how phospholipid structure creates a bilayer.

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Contrast check

- Do not stop at: The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.
- Stronger explanation: Tonicity predicts water movement and cell volume changes.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.

DEEPER

Tonicity predicts water movement and cell volume changes.

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Module 05: Membranes Retrieval and Lab Check

- Module focus: Membranes
- Retrieval prompt: Why do phospholipids form bilayers in water?
- Answer check: Describe how phospholipid structure creates a bilayer.
- Required terms: Phospholipid, Bilayer, Selective permeability
- Lab connection: lab-05_viewing-life.md; lab-05_viewing-life.md supplies evidence for observable boundaries, structures, and environments in living samples.

Module 05 / Retrieval Card

Module 05: Membranes Retrieval and Lab Check

Routine

Cover notes -> answer aloud -> cite evidence -> revise

1 Cover notes | 2 Answer aloud | 3 Cite evidence | 4 Revise

Why do phospholipids form bilayers in water?

Check: Describe how phospholipid structure creates a bilayer.

What does selective permeability mean?

Check: Predict which molecules cross membranes easily.

How does diffusion depend on concentration gradients?

Check: Compare passive and active transport.

How is osmosis different from general diffusion?

Check: Use concentration gradients to explain movement across membranes.

Terms to use

Phospholipid, Bilayer, Selective permeability, Diffusion, Osmosis

Lab connection

lab-05_viewing-life.md supplies evidence for observable boundaries, structures, and environments in living...

Intro biology routine: claim, evidence, reasoning, revision.

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Practice quiz bridge

- 1. Why do phospholipids arrange themselves into a bilayer in water?
- 2. A small nonpolar molecule such as oxygen gas crosses the membrane mainly by:
- 3. An animal cell is placed in pure water, a hypotonic solution. What happens?
- 4. Moving ions against their concentration gradient requires:

QUESTION 1

Answer first

Why do phospholipids arrange themselves into a bilayer in water?

QUESTION 2

Answer first

A small nonpolar molecule such as oxygen gas crosses the membrane mainly by:

QUESTION 3

Answer first

An animal cell is placed in pure water, a hypotonic solution. What happens?

QUESTION 4

Answer first

Moving ions against their concentration gradient requires:

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Synthesis and exit ticket

- Exit claim: explain phospholipid bilayer structure using evidence from lab-05_viewing-life.md.
- Must include: Phospholipid, Bilayer, Selective permeability.
- Revision prompt: what evidence would change your explanation?

CLAIM
Phospholipid bilayer structure

EVIDENCE
lab-05_viewing-life.md

REVISION
What would change your mind?